

Practical No : 1

Terminal 1 :

```
javac *.java
rmiregistry
```

```
bhush@bhush-HP-EliteBook-840-G3: ~/Downloads/DS_Assignment1/Assignment... × bhush@bhush-HP-EliteBook-840-G3: ~/Downloads/DS_Assignment1/Assignment..
bhush@bhush-HP-EliteBook-840-G3:~/Downloads/DS_Assignment1/Assignment-1_LPV$ javac *.java
bhush@bhush-HP-EliteBook-840-G3:~/Downloads/DS_Assignment1/Assignment-1_LPV$ rmiregistry
WARNING: A terminally deprecated method in java.lang.System has been called
WARNING: System::setSecurityManager has been called by sun.rmi.registry.RegistryImpl
WARNING: Please consider reporting this to the maintainers of sun.rmi.registry.RegistryImpl
WARNING: System::setSecurityManager will be removed in a future release
```

Terminal 2 :

```
javac Server.java
java Server
```

```
bhush@bhush-HP-EliteBook-840-G3: ~/Downloads/DS_Assignment1/Assignment... × bhush@bhush-HP-EliteBook-840-G3: ~/Downloads/DS_Assignment1/Assignment..
bhush@bhush-HP-EliteBook-840-G3:~/Downloads/DS_Assignment1/Assignment-1_LPV$ javac server
error: Class names, 'server', are only accepted if annotation processing is explicitly requested
1 error
bhush@bhush-HP-EliteBook-840-G3:~/Downloads/DS_Assignment1/Assignment-1_LPV$ javac Server.java
bhush@bhush-HP-EliteBook-840-G3:~/Downloads/DS_Assignment1/Assignment-1_LPV$ java Server
Server Started....
^Cbhush@bhush-HP-EliteBook-840-G3:~/Downloads/DS_Assignment1/Assignment-1_LPV$ java Server
Server Started....
```

Terminal 3 :

```
java Client
Enter 1st String : abc
Enter 2nd String : xyz
then Output
```

```
bhush@bhush-HP-EliteBook-840-G3: ~/Downloads/DS_Assignment1/Assignment... × bhush@bhush-HP-EliteBook-840-G3: ~/Downloads/DS_Assignment1/Assignment..
bhush@bhush-HP-EliteBook-840-G3:~/Downloads/DS_Assignment1/Assignment-1_LPV$ java Client
Enter First String: Bhushan
Enter Second String: Ahire
----- Results -----
Strings After Joining Is: BhushanAhire
bhush@bhush-HP-EliteBook-840-G3:~/Downloads/DS_Assignment1/Assignment-1_LPV$ ^C
bhush@bhush-HP-EliteBook-840-G3:~/Downloads/DS_Assignment1/Assignment-1_LPV$ java Client
```

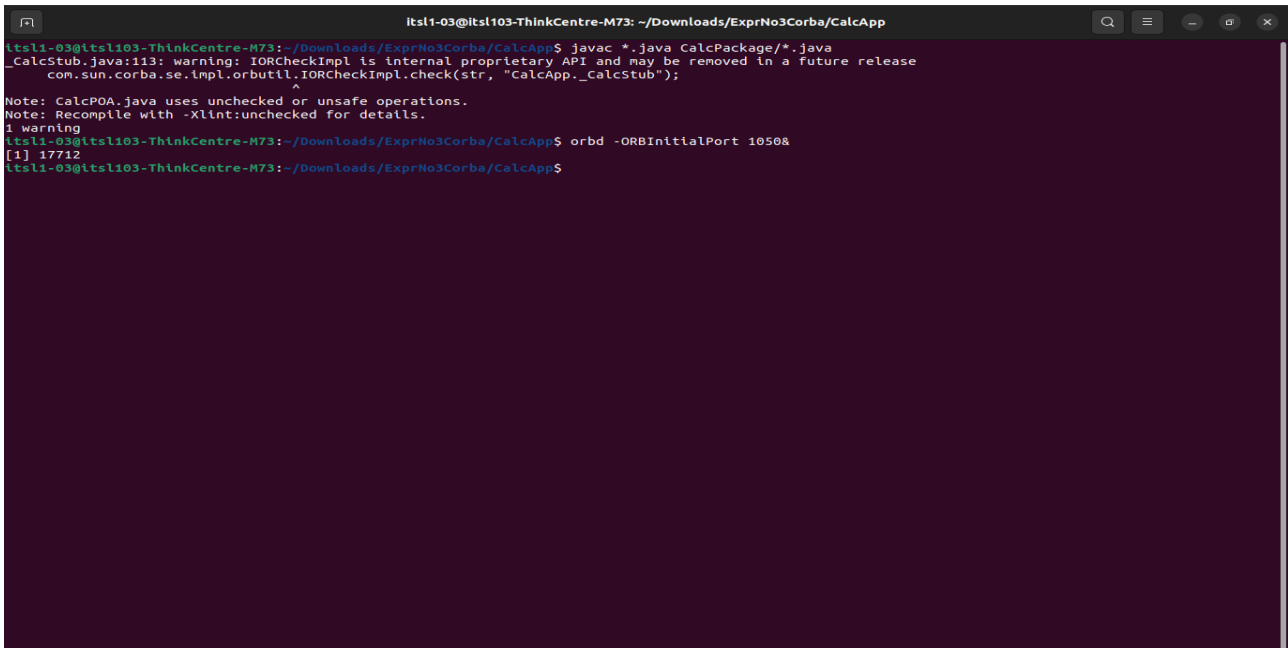
Practical No : 2

Terminal 1:-

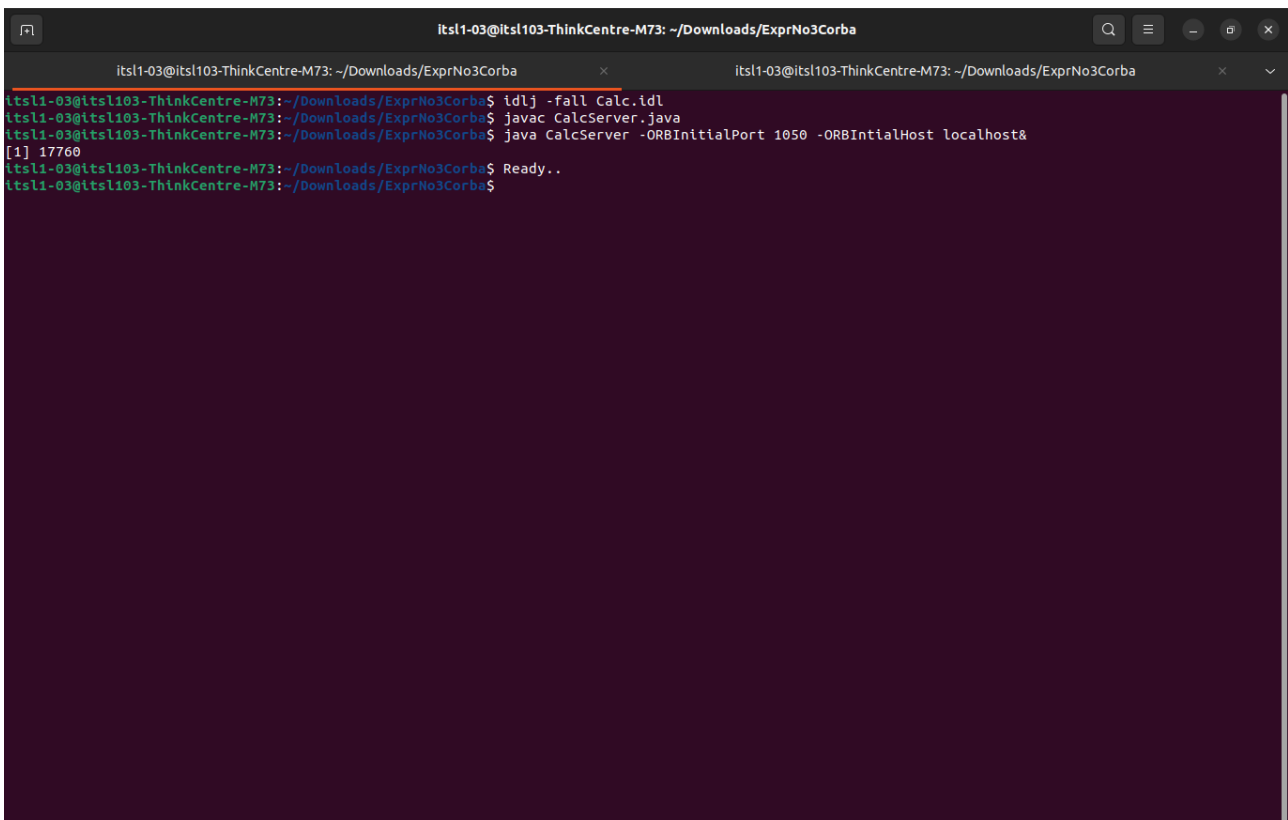
idlj -fall Calc.idl

orbd -ORBInitialPort 1050 &

java CalcServer -ORBInitialPort 1050 -ORBInitialHost localhost



```
itsl1-03@itsl103-ThinkCentre-M73: ~/Downloads/ExprNo3Corba/CalcApp
itsl1-03@itsl103-ThinkCentre-M73:~/Downloads/ExprNo3Corba/CalcApp$ javac *.java CalcPackage/*.java
_CalcStub.java:113: warning: IORCheckImpl is internal proprietary API and may be removed in a future release
    com.sun.corba.se.impl.orbutil.IORCheckImpl.check(str, "CalcApp._CalcStub");
                                ^
Note: CalcPOA.java uses unchecked or unsafe operations.
Note: Recompile with -Xlint:unchecked for details.
1 warning
itsl1-03@itsl103-ThinkCentre-M73:~/Downloads/ExprNo3Corba/CalcApp$ orbd -ORBInitialPort 1050&
[1] 17712
itsl1-03@itsl103-ThinkCentre-M73:~/Downloads/ExprNo3Corba/CalcApp$
```



```
itsl1-03@itsl103-ThinkCentre-M73: ~/Downloads/ExprNo3Corba
itsl1-03@itsl103-ThinkCentre-M73:~/Downloads/ExprNo3Corba$ idlj -fall Calc.idl
itsl1-03@itsl103-ThinkCentre-M73:~/Downloads/ExprNo3Corba$ javac CalcServer.java
itsl1-03@itsl103-ThinkCentre-M73:~/Downloads/ExprNo3Corba$ java CalcServer -ORBInitialPort 1050 -ORBInitialHost localhost&
[1] 17760
itsl1-03@itsl103-ThinkCentre-M73:~/Downloads/ExprNo3Corba$ Ready..
itsl1-03@itsl103-ThinkCentre-M73:~/Downloads/ExprNo3Corba$
```

```

nandini_chavan@Hp: /mnt/c/L X nandini_chavan@Hp: /mnt/c/Usi X nandini_chavan@Hp: /mnt/c/Use X + v
nandini_chavan@Hp:~$ cd /mnt/c/Users/nandi/OneDrive/Desktop/DS_Practicals/DS_Assignment2/ExprNo3Corba
nandini_chavan@Hp: /mnt/c/Users/nandi/OneDrive/Desktop/DS_Practicals/DS_Assignment2/ExprNo3Corba$ idlj -fall Calc.idl
nandini_chavan@Hp: /mnt/c/Users/nandi/OneDrive/Desktop/DS_Practicals/DS_Assignment2/ExprNo3Corba$ javac CalcServer.java
./CalcApp/_CalcStub.java:113: warning: IORCheckImpl is internal proprietary API and may be removed in a future release
    com.sun.corba.se.impl.orbutil.IORCheckImpl.check(str, "CalcApp._CalcStub");
                                ^
Note: ./CalcApp/CalcPOA.java uses unchecked or unsafe operations.
Note: Recompile with -Xlint:unchecked for details.
1 warning
nandini_chavan@Hp: /mnt/c/Users/nandi/OneDrive/Desktop/DS_Practicals/DS_Assignment2/ExprNo3Corba$ orbd -ORBInitialPort 1050 &
[1] 11780
nandini_chavan@Hp: /mnt/c/Users/nandi/OneDrive/Desktop/DS_Practicals/DS_Assignment2/ExprNo3Corba$ May 04, 2026 8:21:42 AM com
.sun.corba.se.impl.transport.SocketOrChannelAcceptorImpl initialize
SEVERE: "IOP00410216: (COMM_FAILURE) Unable to create listener thread on the specified port: 1049"
org.omg.CORBA.COMM_FAILURE: vmcid: SUN minor code: 216 completed: No
    at com.sun.corba.se.impl.logging.ORBUtilSystemException.createListenerFailed(ORBUtilSystemException.java:2632)
    at com.sun.corba.se.impl.logging.ORBUtilSystemException.createListenerFailed(ORBUtilSystemException.java:2651)
    at com.sun.corba.se.impl.transport.SocketOrChannelAcceptorImpl.initialize(SocketOrChannelAcceptorImpl.java:164)
    at com.sun.corba.se.impl.transport.CorbaTransportManagerImpl.getAcceptors(CorbaTransportManagerImpl.java:218)
    at com.sun.corba.se.impl.transport.CorbaTransportManagerImpl.addToIORTemplate(CorbaTransportManagerImpl.java:236)
    at com.sun.corba.se.spi.oa.ObjectAdapterBase.initializeTemplate(ObjectAdapterBase.java:122)
    at com.sun.corba.se.impl.oa.toa.TOAImpL.<init>(TOAImpL.java:96)
    at com.sun.corba.se.impl.oa.toa.TOAFactory.getTOA(TOAFactory.java:90)
    at com.sun.corba.se.impl.orb.ORBImpl.connect(ORBImpl.java:1633)
    at com.sun.corba.se.impl.activation.RepositoryImpl.<init>(RepositoryImpl.java:94)
    at com.sun.corba.se.impl.activation.ORBBD.startActivationObjects(ORBBD.java:274)
    at com.sun.corba.se.impl.activation.ORBBD.run(ORBBD.java:129)
    at com.sun.corba.se.impl.activation.ORBBD.main(ORBBD.java:343)
Caused by: java.net.BindException: Address already in use
    at sun.nio.ch.Net.bind0(Native Method)
    at sun.nio.ch.Net.bind(Net.java:461)

```

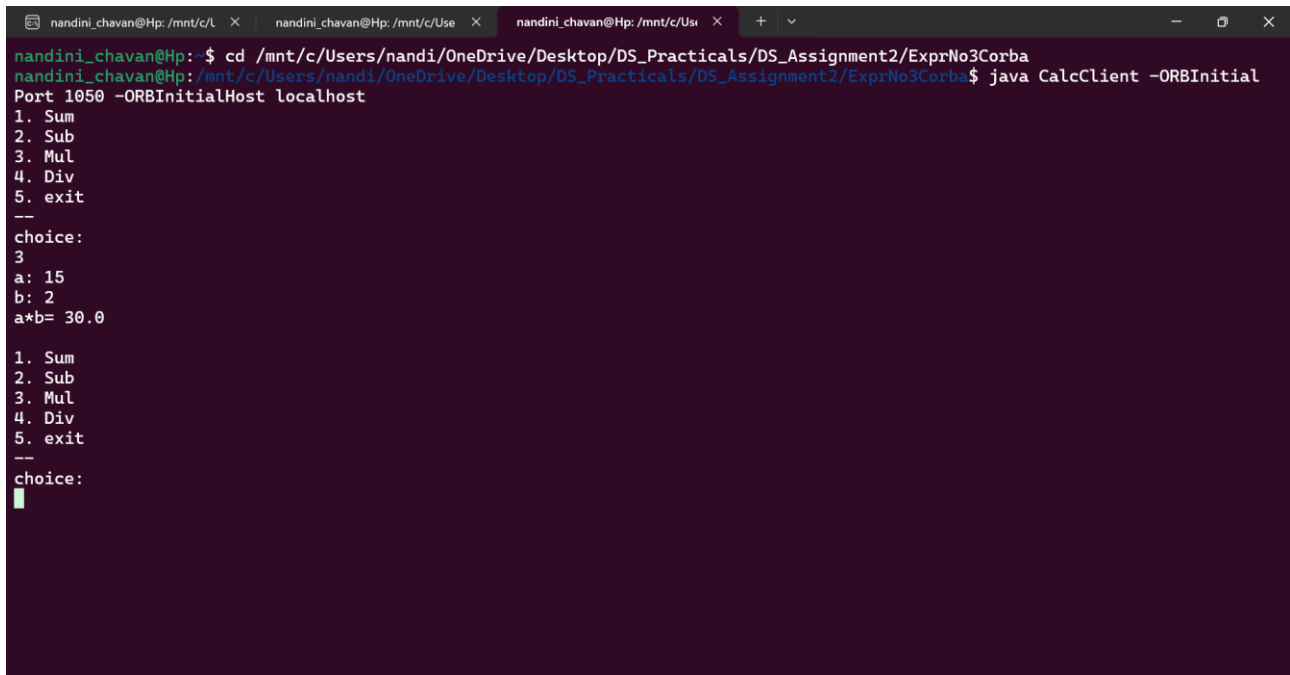
```

    at com.sun.corba.se.impl.activation.RepositoryImpl.<init>(RepositoryImpl.java:94)
    at com.sun.corba.se.impl.activation.ORBBD.startActivationObjects(ORBBD.java:274)
    at com.sun.corba.se.impl.activation.ORBBD.run(ORBBD.java:129)
    at com.sun.corba.se.impl.activation.ORBBD.main(ORBBD.java:343)
Caused by: org.omg.CORBA.COMM_FAILURE: vmcid: SUN minor code: 216 completed: No
    at com.sun.corba.se.impl.logging.ORBUtilSystemException.createListenerFailed(ORBUtilSystemException.java:2632)
    at com.sun.corba.se.impl.logging.ORBUtilSystemException.createListenerFailed(ORBUtilSystemException.java:2651)
    at com.sun.corba.se.impl.transport.SocketOrChannelAcceptorImpl.initialize(SocketOrChannelAcceptorImpl.java:164)
    at com.sun.corba.se.impl.transport.CorbaTransportManagerImpl.getAcceptors(CorbaTransportManagerImpl.java:218)
    at com.sun.corba.se.impl.transport.CorbaTransportManagerImpl.addToIORTemplate(CorbaTransportManagerImpl.java:236)
    at com.sun.corba.se.spi.oa.ObjectAdapterBase.initializeTemplate(ObjectAdapterBase.java:122)
    at com.sun.corba.se.impl.oa.toa.TOAImpL.<init>(TOAImpL.java:96)
    at com.sun.corba.se.impl.oa.toa.TOAFactory.getTOA(TOAFactory.java:90)
    at com.sun.corba.se.impl.orb.ORBImpl.connect(ORBImpl.java:1633)
    ... 4 more
Caused by: java.net.BindException: Address already in use
    at sun.nio.ch.Net.bind0(Native Method)
    at sun.nio.ch.Net.bind(Net.java:461)
    at sun.nio.ch.Net.bind(Net.java:453)
    at sun.nio.ch.ServerSocketChannelImpl.bind(ServerSocketChannelImpl.java:222)
    at sun.nio.ch.ServerSocketAdaptor.bind(ServerSocketAdaptor.java:85)
    at sun.nio.ch.ServerSocketAdaptor.bind(ServerSocketAdaptor.java:78)
    at com.sun.corba.se.impl.transport.DefaultSocketFactoryImpl.createServerSocket(DefaultSocketFactoryImpl.java:83)
    at com.sun.corba.se.impl.transport.SocketOrChannelAcceptorImpl.initialize(SocketOrChannelAcceptorImpl.java:161)
    ... 10 more
^C
[1]+  Done                  orbd -ORBInitialPort 1050
nandini_chavan@Hp: /mnt/c/Users/nandi/OneDrive/Desktop/DS_Practicals/DS_Assignment2/ExprNo3Corba$ java CalcServer -ORBInitial
Port 1050 -ORBInitialHost localhost
Ready..

```

Terminal 2:-

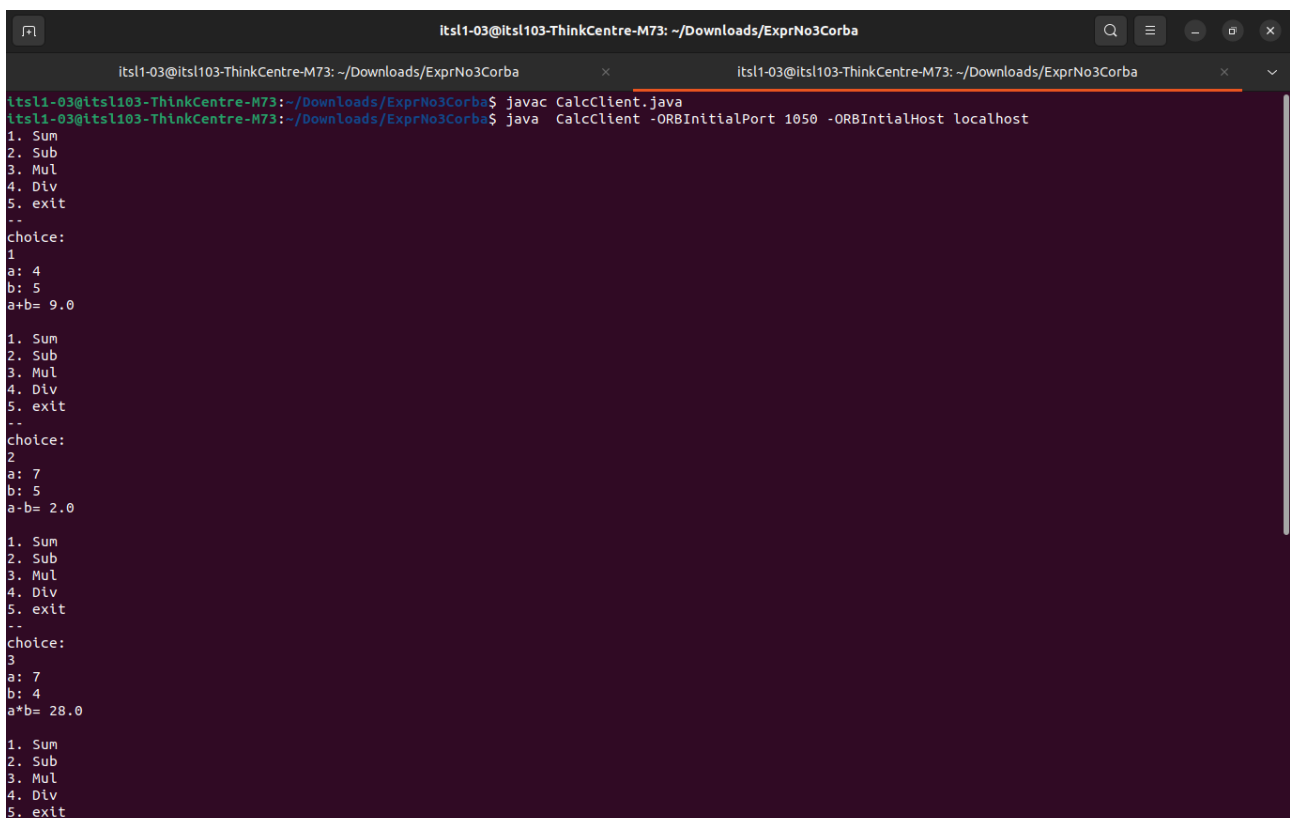
java CalcClient -ORBInitialPort 1050 -ORBInitialHost localhost



A terminal window with three tabs. The active tab shows the command `java CalcClient -ORBInitialPort 1050 -ORBInitialHost localhost` being executed. The program displays a menu with options 1. Sum, 2. Sub, 3. Mul, 4. Div, and 5. exit. The user selects option 3 (Mul), enters values 15 and 2, and the program outputs `a*b= 30.0`. The menu is shown again, and the user's cursor is at the 'choice:' prompt.

```
nandini_chavan@Hp: /mnt/c/L X nandini_chavan@Hp: /mnt/c/Use X nandini_chavan@Hp: /mnt/c/Use X + v
nandini_chavan@Hp:~$ cd /mnt/c/Users/nandi/OneDrive/Desktop/DS_Practicals/DS_Assignment2/ExprNo3Corba
nandini_chavan@Hp: /mnt/c/Users/nandi/OneDrive/Desktop/DS_Practicals/DS_Assignment2/ExprNo3Corba$ java CalcClient -ORBInitial
Port 1050 -ORBInitialHost localhost
1. Sum
2. Sub
3. Mul
4. Div
5. exit
--
choice:
3
a: 15
b: 2
a*b= 30.0

1. Sum
2. Sub
3. Mul
4. Div
5. exit
--
choice:
█
```



A terminal window with two tabs. The active tab shows the command `java CalcClient -ORBInitialPort 1050 -ORBInitialHost localhost` being executed. The program displays a menu with options 1. Sum, 2. Sub, 3. Mul, 4. Div, and 5. exit. The user selects option 1 (Sum), enters values 4 and 5, and the program outputs `a+b= 9.0`. The user then selects option 2 (Sub), enters values 7 and 5, and the program outputs `a-b= 2.0`. Finally, the user selects option 3 (Mul), enters values 7 and 4, and the program outputs `a*b= 28.0`. The menu is shown again, and the user's cursor is at the 'choice:' prompt.

```
itsl1-03@itsl103-ThinkCentre-M73: ~/Downloads/ExprNo3Corba
itsl1-03@itsl103-ThinkCentre-M73: ~/Downloads/ExprNo3Corba$ javac CalcClient.java
itsl1-03@itsl103-ThinkCentre-M73:~/Downloads/ExprNo3Corba$ java CalcClient -ORBInitialPort 1050 -ORBInitialHost localhost
1. Sum
2. Sub
3. Mul
4. Div
5. exit
--
choice:
1
a: 4
b: 5
a+b= 9.0

1. Sum
2. Sub
3. Mul
4. Div
5. exit
--
choice:
2
a: 7
b: 5
a-b= 2.0

1. Sum
2. Sub
3. Mul
4. Div
5. exit
--
choice:
3
a: 7
b: 4
a*b= 28.0

1. Sum
2. Sub
3. Mul
4. Div
5. exit
```

steps to run the program Steps Of Execution :

#Open First Terminal.

idlj -fall Calc.idl //create a calcapp folder

#Now open terminal in calcapp folder.

javac .java CalcPackage/.java

orbd -ORBInitialPort 1050&

#Now go to first terminal where you typed 'idlj -fall Calc.idl' this command.

javac CalcServer.java

java CalcServer -ORBInitialPort 1050 -ORBInitialHost localhost&

#In the same terminal add one more new terminal.

javac CalcClient.java

java CalcClient -ORBInitialPort 1050 -ORBInitialHost localhost

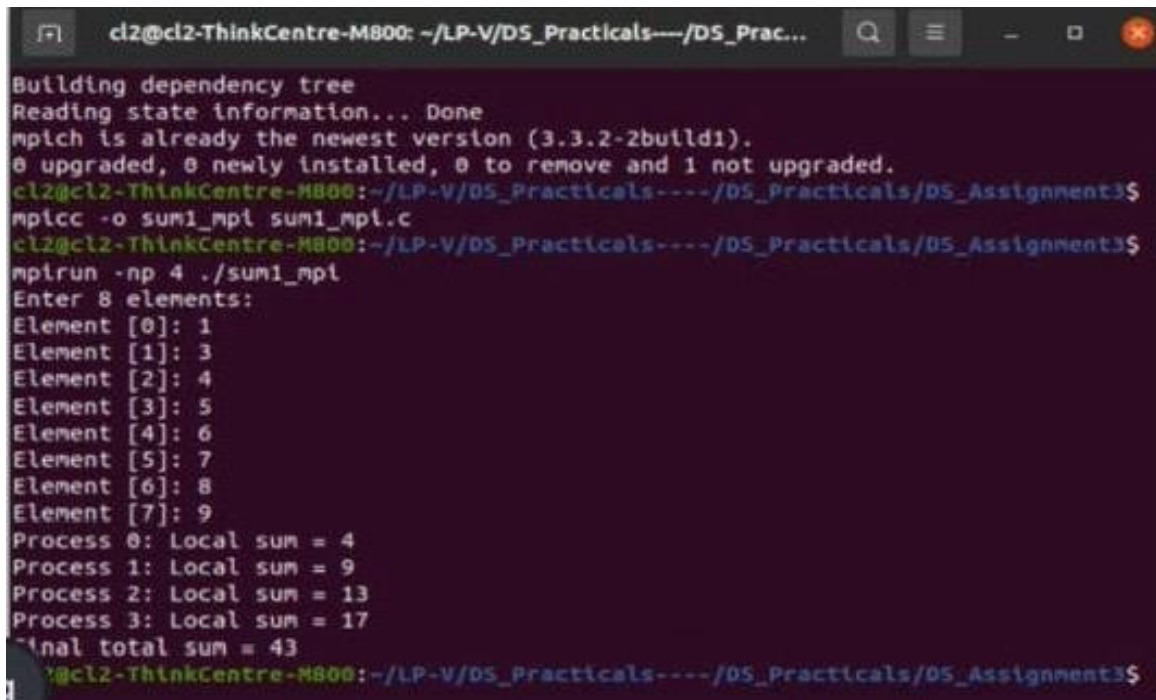
Practical No : 3

installation :

```
sudo apt update  
sudo apt install mpich
```

Terminal 1 :

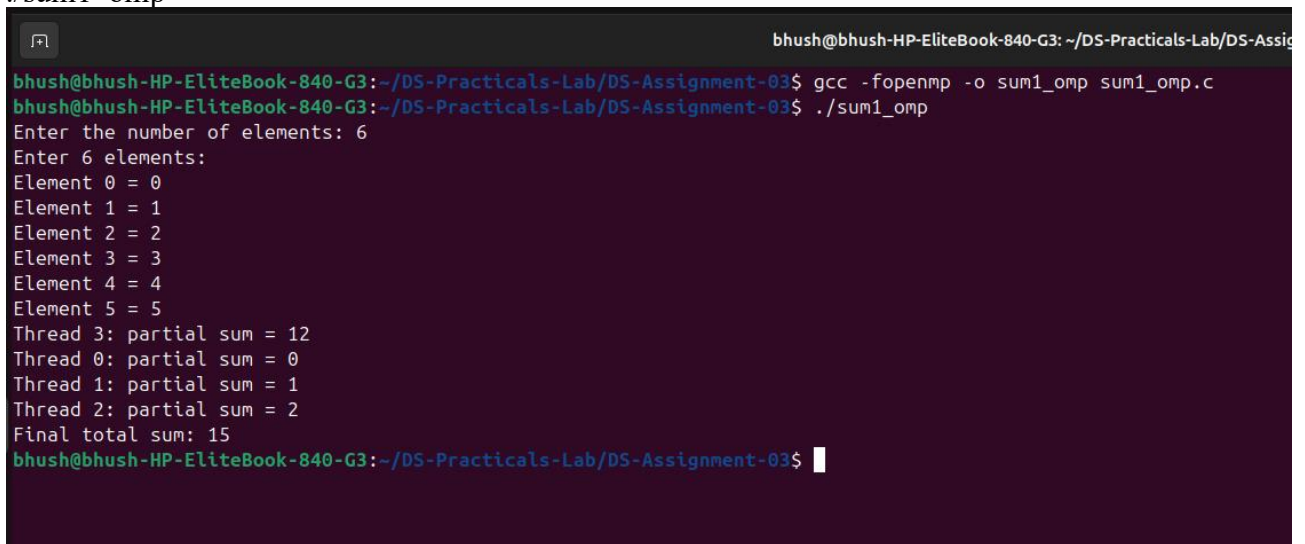
```
mpicc -o sum1_mpi sum1_mpi.c  
mpirun -np 4 ./sum1_mpi
```



```
cl2@cl2-ThinkCentre-M800: ~/LP-V/DS_Practicals---/DS_Prac...  
Building dependency tree  
Reading state information... Done  
mpich is already the newest version (3.3.2-2build1).  
0 upgraded, 0 newly installed, 0 to remove and 1 not upgraded.  
cl2@cl2-ThinkCentre-M800:~/LP-V/DS_Practicals---/DS_Practicals/DS_Assignment3$  
mpicc -o sum1_mpi sum1_mpi.c  
cl2@cl2-ThinkCentre-M800:~/LP-V/DS_Practicals---/DS_Practicals/DS_Assignment3$  
mpirun -np 4 ./sum1_mpi  
Enter 8 elements:  
Element [0]: 1  
Element [1]: 3  
Element [2]: 4  
Element [3]: 5  
Element [4]: 6  
Element [5]: 7  
Element [6]: 8  
Element [7]: 9  
Process 0: Local sum = 4  
Process 1: Local sum = 9  
Process 2: Local sum = 13  
Process 3: Local sum = 17  
Final total sum = 43  
cl2@cl2-ThinkCentre-M800:~/LP-V/DS_Practicals---/DS_Practicals/DS_Assignment3$
```

Terminal 2 :-

```
sudo apt install gcc  
gcc -fopenmp -o sum1_omp sum1_omp.c  
./sum1_omp
```



```
bhush@bhush-HP-EliteBook-840-G3: ~/DS-Practicals-Lab/DS-Assig  
bhush@bhush-HP-EliteBook-840-G3:~/DS-Practicals-Lab/DS-Assignment-03$ gcc -fopenmp -o sum1_omp sum1_omp.c  
bhush@bhush-HP-EliteBook-840-G3:~/DS-Practicals-Lab/DS-Assignment-03$ ./sum1_omp  
Enter the number of elements: 6  
Enter 6 elements:  
Element 0 = 0  
Element 1 = 1  
Element 2 = 2  
Element 3 = 3  
Element 4 = 4  
Element 5 = 5  
Thread 3: partial sum = 12  
Thread 0: partial sum = 0  
Thread 1: partial sum = 1  
Thread 2: partial sum = 2  
Final total sum: 15  
bhush@bhush-HP-EliteBook-840-G3:~/DS-Practicals-Lab/DS-Assignment-03$
```

Practical No : 4

installation :

sudo apt install python-is-python3

Terminal 1 :

python3 server.py

```
bhush@bhush-HP-EliteBook-840-G3:~/Downloads/DS_Practicals/DS_Assignment4$ python3 server.py
Socket at master node created successfully

Clock server started...

Starting to make connections...

Starting synchronization parallely...

New synchronization cycle started.
Number of clients to be synchronized: 0
No client data. Synchronization not applicable.

New synchronization cycle started.
Number of clients to be synchronized: 0
No client data. Synchronization not applicable.

New synchronization cycle started.
Number of clients to be synchronized: 0
No client data. Synchronization not applicable.

New synchronization cycle started.
Number of clients to be synchronized: 0
No client data. Synchronization not applicable.
```

Terminal 2 :

python3 client.py

```
bhush@bhush-HP-EliteBook-840-G3:~/Downloads/DS_Practicals/DS_Assignment4$ python3 client.py
Starting to receive time from server

Starting to receiving synchronized time from server

Recent time sent successfully

Synchronized time at the client is: 2026-05-02 12:35:13.316274

Recent time sent successfully

Synchronized time at the client is: 2026-05-02 12:35:18.317180

Recent time sent successfully

Synchronized time at the client is: 2026-05-02 12:35:23.318605

Recent time sent successfully

Synchronized time at the client is: 2026-05-02 12:35:28.319290

Recent time sent successfully

Synchronized time at the client is: 2026-05-02 12:35:33.319888

Recent time sent successfully

Synchronized time at the client is: 2026-05-02 12:35:38.320849

Recent time sent successfully

Synchronized time at the client is: 2026-05-02 12:35:43.321768

Recent time sent successfully
```

Practical No:5

Practical No : 5

Terminal 1:

```
javac tokenring.java  
java tokenring
```

```
bhush@bhush-HP-EliteBook-840-G3:~/Downloads/DS_Practicals/DS_Assignments$ javac tokenring.java  
bhush@bhush-HP-EliteBook-840-G3:~/Downloads/DS_Practicals/DS_Assignments$ java tokenring  
Enter the num of nodes:  
5  
 0 1 2 3 4 0  
Enter sender:  
1  
Enter receiver:  
4  
Enter Data:  
45  
Token passing: 0-> 1  
Sender 1 sending data: 45  
data 45 forwarded by 2  
data 45 forwarded by 3  
Receiver 4 received data: 45  
  
Do you want to send again?? enter 1 for Yes and 0 for No : 1  
Enter sender:  
2  
Enter receiver:  
2  
Enter Data:  
43  
Token passing: 1-> 2  
Sender 2 sending data: 43  
data 43 forwarded by 3  
data 43 forwarded by 4  
data 43 forwarded by 0  
data 43 forwarded by 1  
Receiver 2 received data: 43  
  
Do you want to send again?? enter 1 for Yes and 0 for No : 0  
bhush@bhush-HP-EliteBook-840-G3:~/Downloads/DS_Practicals/DS_Assignments$
```


Practical No : 6

Terminal 1 :

```
javac Bully.java
java Bully
```

```
bhush@bhush-HP-EliteBook-840-G3: ~/Downloads/DS_Practicals/DS_Assignment6$ javac Bully.java
bhush@bhush-HP-EliteBook-840-G3: ~/Downloads/DS_Practicals/DS_Assignment6$ java Bully
Bully Algorithm
1. Create processes
2. Display processes
3. Up a process
4. Down a process
5. Run election algorithm
6. Exit Program
Enter your choice:- 1
Enter the number of processes:- 4
Creating processes..
P1 created
P2 created
P3 created
P4 created
Process P4 is the coordinator
Bully Algorithm
1. Create processes
2. Display processes
3. Up a process
4. Down a process
5. Run election algorithm
6. Exit Program
Enter your choice:- 2
P1 is up
P2 is up
P3 is up
P4 is up
Process P4 is the coordinator
Bully Algorithm
1. Create processes
2. Display processes
3. Up a process
4. Down a process
5. Run election algorithm
6. Exit Program
Enter your choice:- 3
Enter the process number to up:- 3
Process 3 is already up.
Bully Algorithm
1. Create processes
2. Display processes
3. Up a process
4. Down a process
5. Run election algorithm
6. Exit Program
```

Terminal No : 2

```
javac Ring.java
java Ring
```

```
bhush@bhush-HP-EliteBook-840-G3: ~/Downloads/DS_Practicals/DS_Assignment6$ javac Ring.java
bhush@bhush-HP-EliteBook-840-G3: ~/Downloads/DS_Practicals/DS_Assignment6$ java Ring
Ring Algorithm
1. Create processes
2. Display processes
3. Up a process
4. Down a process
5. Run election algorithm
6. Exit Program
Enter your choice:- 1
Enter the total number of processes:- 4
P1 created.
P2 created.
P3 created.
P4 created.
P4 is the coordinator
Ring Algorithm
1. Create processes
2. Display processes
3. Up a process
4. Down a process
5. Run election algorithm
6. Exit Program
Enter your choice:- 2
P1 is up.
P2 is up.
P3 is up.
P4 is up.
P4 is the coordinator
Ring Algorithm
1. Create processes
2. Display processes
3. Up a process
4. Down a process
5. Run election algorithm
6. Exit Program
Enter your choice:- 3
Enter the process to up:- 2
Process P2 is already up.
Ring Algorithm
1. Create processes
2. Display processes
3. Up a process
4. Down a process
5. Run election algorithm
6. Exit Program
```